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RESEARCH ARTICLE

TiWA: Achieving Tetra Indicator Wi-Fi Associations in Software Defined Wi-Fi Networks

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ABSTRACT With the rise in smart devices, challenges such as traffic management particularly in the indoor scenarios have also increased. Wireless technologies seek to address these needs by using new radio access technologies that offer faster connectivity and access to the Internet. While 4G and 5G provide faster connectivity outdoor, Wi-Fi still remains the dominant technology in indoor scenarios due to its low cost and easy deployment. However, the IEEE 802.11 standard does not address the fairness of load in user-access point (AP) associations, resulting in unequal user distributions and subpar resource utilization. Densification has been employed in the past to address the aforementioned issue, but it is costly as it requires more hardware. Load balancing is necessary in high-density Wi-Fi networks in order to guarantee quality of service (QoS). To avoid the densification and issues related to load balancing at the APs this paper introduces tetra indicator Wi-Fi associations (TiWA), a combined SDN-based user association and channel assignment solution for Wi-Fi networks. TiWA considers multi-criteria to make better association decisions. The wireless stations from an overloaded OpenFlow enabled AP (OAP) are handed to an underloaded OAP by considering multi-metrics such as the packet loss rate, received signal strength indicator (RSSI), throughput and channel occupancy. Extensive experimentations on 5G EmPOWER platform show that TiWA surpasses previous research methods based on load balancing in terms of aggregated throughput by 39%, offering superior level of load fairness and supporting greater numbers of users and traffic prior to hitting the point at which densification is required.

INDEX TERMS Associations, load balancing, multi-criteria, SDN, Wi-Fi.

I. INTRODUCTION

In the driving march towards a wireless society, a growing number of portable gadgets are now incorporating at least one radio access technology (RAT) [1]. 4G and 5G RATs are offering wider geographic coverage and faster connectivity to keep up with the needs of mobile traffic [2]. Wi-Fi, on the other hand, is the favored technology for inside scenarios due to its low price and ease of deployment, despite the indoor signal quality constraints brought on by building materials and distance from base stations [3]. Wi-Fi contention-based design might lead to less than ideal performance. Wi-Fi makes use of unlicensed frequency bands and there are fewer channels like the 2.4 GHz band available making it more susceptible to interference from co-located deployments. As a result, escaping collision domains is challenging [4].

When a large number of APs share the same collision domain, it can result in a degraded throughput, particularly when the ratio of APs to units of the area rises [5]. In scenarios where a single AP cannot meet the needs of every single user, densification is often employed as a solution [6]. More APs are deployed and transmission power is reduced to alleviate the load on overloaded APs as depicted in Figure 1. However, densification can exacerbate the problems mentioned above,

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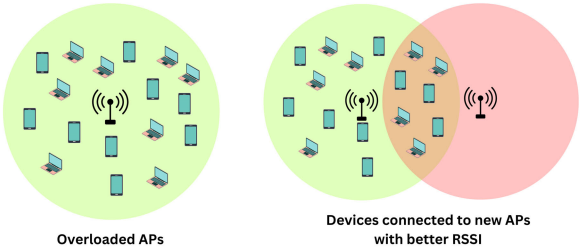


FIGURE 1. Densification.

as well as increase the infrastructure costs, making it necessary to explore more revolutionary approaches to address this problem.

Policies for managing users and resources at the air interface might be ineffective irrespective of the density level because of the absence of standard methods for user-AP association [7]. The decision regarding the association method is left to the vendor as the IEEE 802.11 standard does not mandate any particular methodology [8]. Most vendors follow a client-driven (CD) approach [9], which is also known as a scheme based on RSSI in the literature, for the sake of simplicity.

In the RSSI based scheme, the devices opt for the AP that exhibits the utmost RSSI. The downside of relying solely on RSSI is that it overlooks critical details about the disturbances and obstructions present in the wireless channel and network load [10]. Consequently, there is a risk of uneven distribution of the load across the infrastructure, which can result in suboptimal allocation of network resources [11].

The aforementioned issue is even worse in the situations with large user concentrations brought on by flash events, such as boarding calls at airport gates or autograph sessions in malls, where every wireless device tries to connect to only certain APs, leaving the rest APs underloaded [12]. In several environments, including office buildings and airports, the distribution of users is unequal [13]. As a result, certain APs become overloaded while others are free. Densification is a part of conventional resolution techniques for overloaded APs. Although the network load distribution is unequal, there are resources that might be used to prevent the requirement for more equipment to be installed [14]. Rearranging the user distribution map to divert some of the traffic from the overloaded APs and using available network resources wherever possible is a more effective strategy [15]. This approach would eliminate the need for expensive and challenging-to-manage densification solutions and would be applicable to scenarios where there is a significant rise in the user count throughout the network.

In recent years SDN has become an innovative approach for revamping network services [16].SDN used in many IoT frameworks [17] and smart contract solution [18], but was not much extended for Wi-Fi networks in past studies which was very challenging to use SDN for Wi-Fi networks. By supplying high-level programming abstractions, SDN separates the control plane from the data plane and enables a centralized network management. The controller is placed in the control plane enabling more complex and accurate management activities with a two-way information flow. The controller may collect real-time data from the network devices, which is essential for promptly taking a variety of important network wide decisions like managing mobility and identifying and fixing defects. In wired networks, OpenFlow acts as a de facto standard for SDN, while no comparable dominating framework exists for wireless networks [19]. 5G-EmPOWER and Odin are two potential frameworks for wireless networks [20].

In conventional IEEE 802.11 WLANs, the APs form the network operate without the coordination, therefore information about the load on each AP and the resources that are available to it is not shared among them [21]. On the other hand, a centralized association scheme with a thorough understanding of the network, like that offered by an SDN-controller, can make association decisions more effectively [22]. In cases of unequal user distribution, SDN proactive network reaction capability allows it to shift wireless devices from high-traffic APs to low-traffic APs neighbours [23]. As a result, there is no need to build new infrastructure because the wireless devices can utilize resources that are currently not in use. The traffic handled by one AP in the shared domain affects that of the other, making channel occupancy an essential factor to consider [24]. The strength of the signal is also important since relocating a wireless device too far causes more retransmissions and an modulation and coding scheme (MCS) with a lower physical rate keeps the channel busy for a longer time and wastes resources [8].

The presented work introduces a Tetra indicator Wi-Fi Association (TiWA), which is a user association scheme that aims to improve the utilization of available wireless resources, thereby eliminating the need for densification to maintain network performance. TiWA improve channel load estimation based on transmission rates and transmitted data at regular intervals. It also takes into account the four main metrics that influence densification such as packet loss rate, RSSI, throughput and channel occupancy in the wireless resources (i.e, airtime). By considering all these factors, TiWA improve the network resource allocation efficiently. TiWA is mainly based on the mobility of devices and the main challenge that arises to us is that it is very difficult to conduct experiments in an indoor environment due to the restricted mobility of wireless devices due to congested places. We have used three non-overlapping channels in the 2.4 GHz band. However, the load balancing problem becomes significantly more complex in the 5 GHz band. We are considering to use for 5 GHz in future research too. To demonstrate the effectiveness of TiWA, the performance analysis in this research work considers multiple conditions that could lead to densification. The results show that TiWA outperforms state-of-the-art approaches and it can effectively avoid congestion in the wireless medium. Overall, TiWA provides a

promising solution for improving network resource allocation and avoiding the need for densification in wireless networks.

The main contribution of this study is as follow:

- We propose TiWA, a novel user association scheme in Wi-Fi networks that aims to enhance resource usage, avoid densification, and enhance overall network performance. The channel load is estimated on transmission rates and transmitted data.
- TiWA considers four factors such as packet loss rate, RSSI, throughput, and channel occupancy which helps to overcome densification.
- We propose a testbed design for TiWA that verifies the authenticity of the proposed algorithms.

The subsequent sections of the written paper are arranged as follows. Section II offers an outline of the prior work related to the proposed research. Section III explains the principles of TiWA and its design. Section IV presents the experimental setup. Section V examines the results derived from the actual performance evaluation. Lastly, in Section VI, we present our conclusions.

II. RELATED WORK

There has been considerable academic research on linking strategies in wireless local area networks (WLANs) supported by the existing literature. Numerous studies that compare commercial WLAN and Citizens Broadband Radio Service (CBRS) systems have been undertaken [25], [26]. With the emergence of SDN in recent years, more efficient approaches have become possible without requiring modifications to the IEEE 802.11 standards. In this section, we provide an overview of the most significant related works that are categorized by the approach they follow.

A. BANDWIDTH REQUIREMENTS

The studies grouped in this category focus on bandwidth requirements and utilize this information as their primary input. For example, one solution aimed to distribute the load across the APs in the network to achieve fairness in user bandwidth [27]. However, this method requires each wireless device to specify its requirements of bandwidth for the ongoing session beforehand. An algorithm is designed to reduce the number of wireless devices per AP based on how many wireless devices each AP controls and their bandwidth requirements [28]. However, the number of associated wireless devices does not always accurately represent the APs load. The method may result in ping-pong effects, which can be challenging to operate without AP synchronization. The authors address load imbalance issue by implementing a handover only after the same AP has been identified as the best choice for ''n'' number of consecutive times [29]. Another approach aims to maximize the bandwidth allocated to each connected wireless device while maintaining a balanced load [30]. Bandwidth distribution is also a key consideration for AP load balancing where differences in the bandwidth demands of different wireless devices are taken into account [31]. However, these methods only aim to enhance one parameter and ignore channel circumstances.

B. THROUGHPUT MAXIMIZATION

Maximizing the network's aggregate throughput is a common tactic used in various research investigations. Despite the fact that there are many alternate distributions that may be available and may be able to provide superior performance, wireless devices typically connect using a static distribution [32]. A load-balancing algorithm is designed that finds the optimum route for all flows [33]. The algorithm chooses several transmission channels for each information exchange, based on the required QoS. To improve the outcomes, some approaches utilize the concept of SDN principles [34]. The authors make use of graph theory to maximize the throughput by identifying the optimum association graph between wireless devices and APs [35]. All APs must work on the same channel to maintain the required throughput. All virtual resources are abstracted into a single entity referred to as the virtual resource chain (VRC) in order to maximize throughput [36]. Despite concentrating on a single parameter, the throughput maximization technique does not deal with the issue of resources being wasted owing to interference or collisions.

C. DISTRIBUTED APPROACHES

Distributed approaches were popular before the rise of SDN because there was no central controller to provide a comprehensive network view and provision for programmability. This led some researchers to adopt distributed approaches. The authors suggest a system that select the destination AP for the wireless device side by creating new key performance indicators that consider both signal strength and the performance degradation experienced by other wireless devices [37]. A strategy is presented where decisions are made on the wireless device side and are dependent on signal strength [38]. A distributed technique that measures the burden in terms of missed deadlines is also suggested [39]. Wireless devices are then reassigned to another AP when a certain number of deadlines are missed. These distributed techniques are not ideal since they do not take into account network-wide performance, despite the fact that they outperform the conventional wireless device driven association schemes. Since distributed approaches lack a comprehensive understanding of the network, their choices may enhance performance in one AP while impairing that of nearby APs.

D. AP-LEVEL LOAD BALANCING

In the literature, load balancing at the AP level is a method commonly mentioned for distributing traffic throughout the network. High-density Wi-Fi networks require load balancing to efficiently share resources and distribute network traffic to maintain quality of Service (QoS). It maintains a fair allocation of bandwidth, avoids access point overload and congestion, and optimizes performance. In addition to optimizing

seamless roaming, load balancing increases overall network capacity and raises user experience for users in populated locations. The authors use the networks average workload to reroute traffic when a new wireless device joins, however the standard beacon frames need to be modified [40]. As a result, the majority of methods that employ this tactic benefit from SDN, which enables network programmability and the use of light virtual access points (LVAPs). The load at the AP level is balanced by the authors using an SDN centralised controller to complete a scoring matrix that accounts for both RSSI and occupancy rate [41]. The impact of a clever AP selection approach is studied using a fittingness factor parameter [42]. A reassociation technique is employed to improve QoS and decrease interpacket delays using a Markovian analytical model [43]. This method restricts user mobility. Handovers from the APs with the highest load to the APs with the lowest load are made to balance AP load [44]. The scheme does not reduce densification.

Anytime a status change takes place, an SDN controller dispatches wireless devices from overloaded APs to underloaded ones [45]. However, because a number of other factors affect channel conditions, focusing simply on each APs load is not the best approach. Enterprise wireless local area networks (WLANs), are typically implemented in dense networks like those of office buildings or universities, have more severe channel and AP congestion issues [46]. A solution for WLANs that accounts for the number of users per AP and the channel load at the AP level is presented [47]. The channel could be overloaded by more APs in the same collision domain, which would reduce throughput for both APs. For association decisions, the authors incorporate even more criteria, such as RSSI, possible capacity, attainable data rate and users' locations [48]. In dense networks, APs share collision domains, which causes traffic on one AP to effect traffic on the other [49]. As a result, channel resources (airtime) become the bottleneck. Relying exclusively on the load of each AP is not optimal.

E. SOLUTION TO PREVIOUS PROBLEMS

The paper introduces TiWA, a solution that addresses several issues by striking a balance between AP load, channel load, and signal quality, while also remaining compatible with readily available wireless devices. TiWA reduces collision domains and considers channel airtime instead of just individual AP airtime to prevent channel congestion problems, even in unavoidable collision scenarios. Additionally, it uses RSSI levels to maintain an acceptable physical rate, making it more adaptable and effective in various situations. TiWA is designed for practical use in densification scenarios, enabling more users and traffic to use the same resources, thereby delaying the need for densification to handle a substantial increase in users. This study surpasses the current state of the art in various ways, including:

1) TiWA decreases collision domains and integrates four factors to enhance its adaptability beyond works that concentrate solely on bandwidth requirements or maximizing throughput. TiWA employs the available resources more efficiently and recovers previously wasted resources due to collisions.
2) TiWA utilizes SDN principles to program the network without modifying the standard. This gives TiWA a broad perspective of the network and allows it to make better reassociation decisions than distributed approaches.
3) TiWA maintains a constant transmission power and maximizes existing resources to prevent wireless devices from being left out on the edges of the network, as is typical with cell breathing approaches.
4) To avoid performance drops, seamless reassociations are necessary. TiWA employs seamless-handover techniques like LVAPs to reassociate wireless devices and attain much better resource sharing.
5) TiWA avoids channel congestion when collision domains are unavoidable by minimising collision domains and taking into consideration the available airtime in the channel rather than only the airtime of a single AP, which handles load balancing challenges at the AP level.

III. TiWA:TETRA INDICATOR Wi-Fi ASSOCIATION

The focus of this work is the introduction of Wi-Fi Multi-indicator Wireless devices Association (TiWA), which defines four indicators for monitoring resource allocation in a network as shown in Figure 2. The aim is to prevent densification, which occurs when more APs are required to handle network traffic despite the existence of free resources in neighboring areas and load balancing between APs. The objective of TiWA is to study new user distributions that optimize resource utilization. For each AP, the SDN controller collects uplink RSSI and channel utilization data. Based on these metrics, TiWA defines four indicators such as packet loss rate, average RSSI of an AP, throughput and channel occupancy. Through wireless device reassignment, TiWA seeks to distribute users in a way that maximizes the use of available resources and avoids the need for additional infrastructure, thus preventing the problems associated with densification. When wireless devices initially want to connect to APs in a network, the process of AP discovery and selection takes place. APs send the beacon frames. The wireless devices scan for APs by listening for those beacon frames that contain network information, including the RSSI which indicates the signal quality between the device and each AP. The device initiates a prompt handshake with the AP having the strongest RSSI value that gives better signal strength and higher data rates. Once the user chooses a network, the device sends an association request to the selected AP including details like its MAC address and capabilities.

A. INDICATOR 1: APs AVERAGE RSSI

The average RSSI of an AP points to the average of the uplink RSSIs of all the wireless devices associated to the AP.

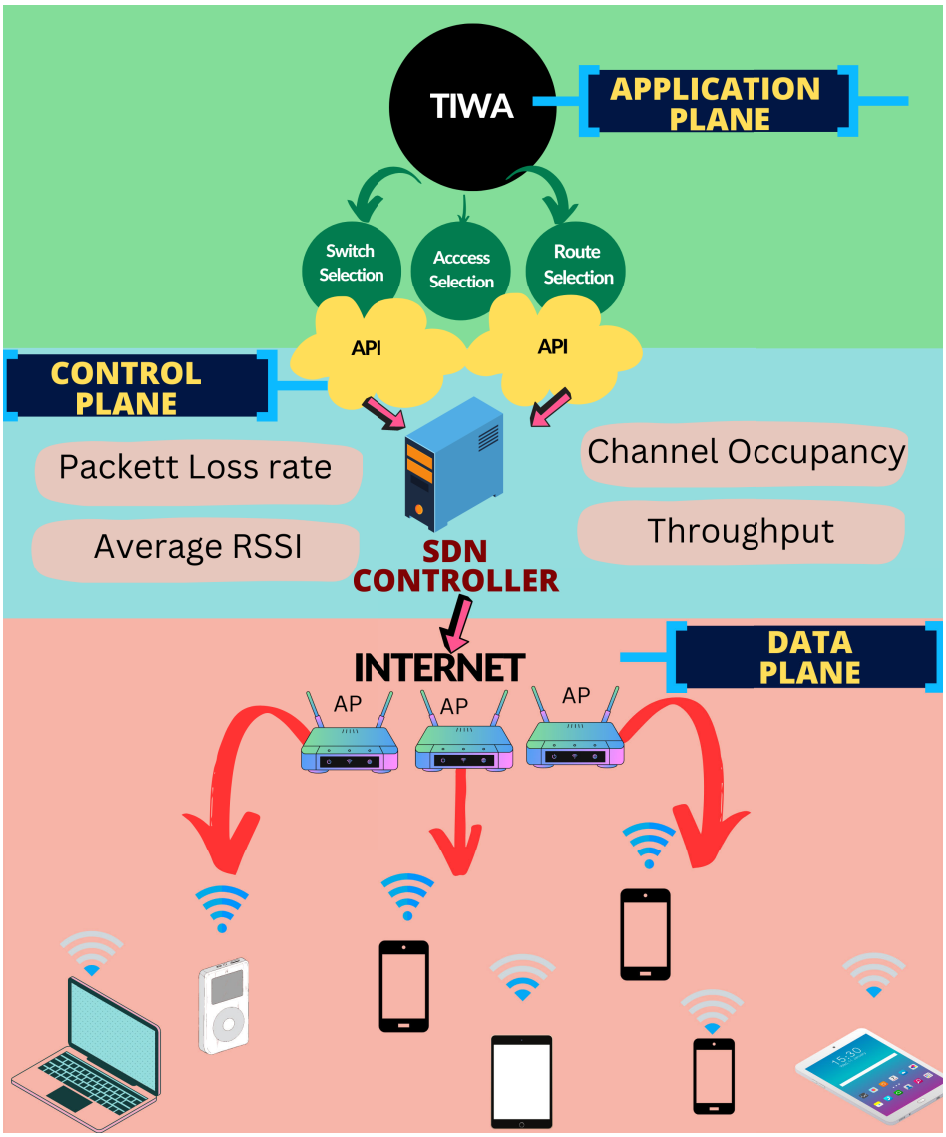


FIGURE 2. The proposed architecture for TiWA.

TABLE 1. Indicators in TiWA.

Indicator	Definition	Measurement	Implication
Average RSSI	Received Signal Strength Indicator	dBm	Stronger signal indicates better connection quality
Packet loss rate	Ratio of lost packets to sent packets	%	Lower value indicates better data transmission
Throughput	Data transfer rate	Mbps	Higher value indicates faster data transmission
Channel Occupancy	Percentage of time the channel is occupied	%	Lower value indicates less congestion

A low average RSSI indicates poor signal strength, which can result in lower physical rates. This lead to more robust signals that waste air time, as transmitted frames may need to be retransmitted due to poor signal strength. Moreover, the average RSSI of an AP may become very low after the redistribution of wireless devices due to the AP load indicator which can lead to increased frame retransmissions. To prevent this, TiWA monitors the average RSSI of an AP and seeks to improve the distribution of wireless devices. TiWA also assesses if redistribution is a viable option upon individual degradation of a wireless devices RSSI, especially for moving wireless devices, to improve overall performance.

Algorithm 1 Wireless Device Association Algorithm

Input:
Average RSSI: the average received signal strength indication (RSSI) at each access point (AP).
Packet Loss Rate: the percentage of packets that do not reach their destination.
Channel Occupancy: the information of channels currently in use at each AP.
Throughput: Network throughput measurement.
Output:
The destination AP to which wireless devices should be associated.
if $(AP_{max} - AP_{min}) > (AP_{med})$ **then**
    For each individual AP and every wireless device, select the destination AP as $Max_j[RSSI_j * (Channelload_{AP_i} + AP_iLoad + Plrate + Throughput)]$
else
    For each individual AP and every wireless device, select the destination AP as $Max_j[RSSI_j * (Channelload_{AP_i} + AP_iLoad + Plrate + Throughput)]$
end if
if $(ChannelOccupancy_{max} - ChannelOccupancy_{min}) > (ChannelOccupancy_{med})$ **then**
    Re-assign the channel.
end if

Algorithm 1 shows a simplified version of the wireless device association. Each of the above-mentioned indications has a corresponding threshold that lines up with the algorithm's if statements. Whenever the $(AP_{max} - AP_{min}) > (AP_{med})$) criterion is met, the updated values are periodically delivered from an AP to the controller. At the controller end, the time frame is programmable. The maximum, median, and minimum statistical are not measured using the immediate values supplied by the APs, which is another crucial component. The historical average of the most recent 10 reported values is applied. A single report's modifications do not have a substantial impact to produce ping-pong effects, thus this is done to prevent them. The first and second signs set off a reassociation procedure that chooses the ideal balance between the amount of network resources consumed and the signal quality. With i denoting the AP and j denoting the wireless device, the reassociation of wireless devices to APs maximize $Max_j[RSSI_j * (Channelload_{AP_i} + AP_iLoad + Plrate + Throughput)]$. RSSI has a negative value, which is the reason why the product is maximized rather than minimized. A higher MCS can be chosen with the aid of good signal quality, which cuts down the time it takes for each frame to be broadcasted. As a result, the channel remains vacant for more extended time and can carry more frames. Having more active APs and channels also lowers the perceived product value. The algorithm therefore looks for the ideal fusion of AP load and the channel load. There won't be a migration if there isn't a better option. It is possible to simplify this issue to the generalized assignment problem (GAP). There are n tasks and m agents in this problem. Each job could be delegated precisely to one agent, yet one agent can be given more than one duty. The agent-task assignment determines the task cost. Finding the collection of agent-task assignments that maximizes profit is the objective. Similar to TiWA, where APs are agents and wireless devices are tasks. TiWA searches for the collection of AP-wireless device assignments that maximizes $Max_j[RSSI_j * (Channelload_{AP_i} + AP_iLoad + Plrate + Throughput)]$. Every wireless device can only have one AP allocated to it, however an AP can have several wireless devices.

B. INDICATOR 2: PACKET LOSS RATE

It measures the percentage of packets lost by an AP during the transmission process. This indicator is important because packet loss can occur due to various factors, such as interference, congestion, or poor signal quality, which can adversely impact the network's performance. TiWA estimates the packet loss rate for each wireless device using the Additive Increase Multiplicative Decrease Rate Control algorithm (ARF). ARF is a feedback-based algorithm that adjusts the data transmission rate of a wireless device based on the packet loss rate. TiWA periodically collects the packet loss rate and the data transmission rate of each wireless device and uses this information to estimate the packet loss rate of the AP. If the packet loss rate of an AP exceeds a certain threshold, TiWA considers this as a congested state and seeks to redistribute wireless devices to less congested APs. This redistribution process is similar to the one used in Indicator 1. In summary, the Packet Loss Rate indicator monitors the quality of the wireless link by estimating the percentage of lost packets during transmission. When the packet loss rate of an AP exceeds a threshold, TiWA seeks to redistribute wireless devices to improve network performance.

Algorithm 2 is used to reassign a channel when the indicator which is higher than the threshold. With the channel reassignment, a channel is recursively assigned to the AP with the fewest available channels. The channels that are open are those that have not yet been utilized by any of the neighboring APs. To minimize the overlapping and interference issues, the algorithm selects the least-used channels when they are available. TiWA removes a channel from the list of accessible channels if it exceeds the threshold, when the total channel load exceeds the threshold. If the channel load is not taken into consideration, a different channel distribution may occur with one of the channels still being overloaded. The adjustment may prevent collision domains from being small. However, having multiple APs with low loads in the same collision domain is better to have a channel that is congested.

C. INDICATOR 3: THROUGHPUT

Throughput is an essential metric in Wi-Fi networks that refers to the volume of data that can be transmitted by a network over a specific duration of time. It is usually measured in bits per second (bps) or megabits per second (Mbps).

Algorithm 2 Channel Management Algorithm

Input:
Average Received Signal Strength Indicator (RSSI)
Channel Utilization
Packet Loss Rate
Throughput
Output:
Sets of Access Points (APs) and Channels
Channel Reassignment
Check for remaining unassigned APs
if All other APs have been assigned **then**
    Proceed to end assignment
else
    End assignment
    Arrange remaining unassigned APs in order of highest number of neighboring APs and lowest channel utilization
    Select a suitable AP from the remaining unassigned APs
    Choose an appropriate channel from the available channels
    Remove any overlapped channels
end if
if No channels are available **then**
    Resort to minimum assignment
    Assign the least loaded channels to the AP
else
    Repeat channel assignment for the next unassigned AP.
end if

Using TiWA, the throughput indicator is used to evaluate the amount of traffic that each AP is handling. When an AP is overloaded with traffic, the throughput decreases, which negatively affects the overall performance of the network. To prevent this, TiWA periodically checks the throughput of each AP and triggers an action when it exceeds a certain threshold. TiWA uses the maximum and minimum throughput values reported by an AP in the past 10 reports to calculate the median throughput value. If the difference between the maximum and minimum values is greater than the median, the throughput indicator triggers an action. This action can include migrating some wireless devices to less loaded APs, redistributing channels, or taking no action at all if there is no better option. By doing this, TiWA can effectively balance the traffic load across the network and ensure that no single AP is overloaded, which can result in a decrease in throughput and overall network performance.

D. INDICATOR 4: CHANNEL OCCUPANCY

This indicator measures the percentage of time that the wireless medium is being used by the AP. When this value is high, it means that the medium is congested and there is a high possibility of collisions and packet loss, which can degrade the overall network performance. To calculate the channel occupancy, TiWA estimates the number of bytes that are being transmitted on each channel, based on the reported delivery probabilities and MCS rates of each wireless device. The channel occupancy is then calculated as the sum of the estimated transmission times of all wireless devices, divided by the total time. If the channel occupancy exceeds a certain threshold, TiWA considers it as a sign of congestion and tries to improve the situation by redistributing wireless devices to other APs with lower channel occupancies. This is done using the same process as in the previous indicators by evaluating the potential benefits of different wireless device redistributions and selecting the one that improves the overall network performance.

Algorithm 1 has a complexity of $O(x * y)$, where x number of APs' and y is the number of wireless devices, as all the wireless devices search through all the possible APs in range to find the best match. This process is repeated by considering RSSI values but to avoid an increase in the complexity this Algorithm changes the order of APs according to $Max_j[RSSI_j * (Channelload_{AP_i} + AP_iLoad + Plrate + Throughput)]$, this recursively call the algorithm 2. As a result, overall complexity become $O(x^2)$.

Algorithm 2 is a recursive approach that ends when all APs have an assignation of channels. The remaining APs are first sorted on each recursive level with attention being given to the one with the fewest available channels and the greatest number of neighbors. Considering x as the number of APs, Then the complexity of this is $O(x \cdot log(x))$. After then, the overlapped channels are removed from the entire list of APs in order to find the available channels. This part has the complexity of $O(x)$. The algorithm then iterates through a list of potential channels that can only be as long as x, hence the complexity is also $O(x)$. Therefore, each recursive call requires $O(x)$ time. But each inner call takes this amount of time. This function is called $(x - 1)$ recursively. Therefore, it concludes $T(x) = T(x - 1) + O(x)$. So, the complexity of this algorithm is $O(x^2)$.

IV. EXPERIMENTAL SETUP

The 5G EmPOWER framework served as a model for the proposed scheme implementation. To support lightweight visualization and heterogeneous RATs like Wi-Fi and LTE, 5G EmPOWER platform combines SDN and NFV. The platform provides complete network state visibility, allowing for the establishment of the network service and quick application prototyping. The study solution is implemented as an application network, giving it a superior position above the controller and allowing it to access real-time network data. Since each AP in the network keeps track of its current status, even if the central controller isn't available, the network can still function.

SDN offers a number of benefits, including the simplicity with which necessary network metrics may be gathered and the capacity to modify network behavior in response to changes in metrics. The controller can configure the frequency of AP reporting regarding channel usage, wireless

devices and received RSSI among other data. However, the data provided by APs is not utilized the right away. In order to avoid unintentional handoffs brought on by burst traffic or other causes that could result in ping-pong effects that would later need to be reversed, the past ten reports are instead stored, and an average is calculated. It is ensured that any changes in channel occupancy or AP load will have the same impact over time by using the average of the last ten reports. This implies that reassociation will only be triggered by major influences that substantially change the average. Additionally, 5G-EmPOWER provides LVAPs, which streamline wireless device state management through a high-level interface. Since every wireless device has a distinct LVAP, even APs that operate on various channels can seamlessly switch over to one another. The IEEE 802.11 standard channel switch announcements (CSAs) are used in these circumstances to instruct the wireless device to start broadcasting on the new channel.

In order to guarantee that beacons and CSA frames are provided to each user in unicast mode, each LVAP has a distinct basic service set identifier (BSSID). This makes it possible for authentication and association to continue working after a handoff. At the destination access point, an LVAP instance is created as part of the handover procedure; this instance is inactive until the wireless device connects to it. A crucial aspect of the suggested method is, after reassociation, it manages infrastructure and keeps performance stable. Handoffs occur seamlessly. Although the method is used in this study for up-link broadcasts, it might also be used for down-link transmissions with IEEE 802.11 k standard. With the exception of selected Apple gadgets, not many devices are capable of using this standard.

V. EVALUATION METHODOLOGY

The performance evaluation aims to show how well the four indicators added to TiWA for improved resource allocation to postpone densification, work. Three experiments make up the evaluation, each of which focuses on a different indicator listed in Section III. These tests entail a deployment that could result in densification. The technique used is intended to lessen the already significant variability in wireless networks that results from analyzing all the indicators in the same trial. It is intended to separate out the effects of each indication so as to arrive at more accurate conclusions may be made for each. Three access points constructed on PCEngines ALIX 2D boards and each outfitted with a single Atheros AR9220 Wireless Network Interface Controller (NIC) served as the actual testbed for the evaluation. IEEE 802.11 n standard was used for the studies, and version 17.01 of EmPOWER-LEDE, an open-source Linux distribution built on OpenWRT, was flashed onto the access points.

The APs were linked to a laptop running Linux Mint 19.2 Tina, which functioned as the 5G-EmPOWER controller, and placed in a star topology. 15 Raspberry Pi, 4 computers running Raspbian Buster with 1 GB of RAM were initially utilized as the wireless devices, but this was changed to a TP-Link TL-WN722N with a 4 dBi omni directional antenna because of the large amount of retransmissions. As mobile wireless devices, two Toshiba laptops running Ubuntu 19.04 and equipped with an Intel Atom processor and 4 GB of RAM were employed. Three other laptops were configured in monitor mode, one for each channel, to collect the headers of all the frames on channel. The core performance indicators were assessed using the headers. The location of the trials was chosen precisely to prevent external network disturbance. The APs were positioned to provide smooth roaming between overlapping service zones. Two wireless devices were placed within each AP's range to watch how reassociations were conducted, while the remaining wireless devices were positioned in the middle of all the APs ranges. Each experiment had unique characteristics that are further discussed in the following subsections, such as the addition of five additional wireless devices, the channels used, a lot of wireless devices producing traffic for each trial component, the number of mobile wireless devices, and the length of each experiment (in cases where the experiment had two aspects) are all listed in Table 2 along with other pertinent data.

For all the experiments, each wireless device is set to transmit at a rate of 3 Mbps to the server located in the controller, resulting in a total network traffic of 30 Mbps.Traffic is generated by the Distributed Internet Traffic Generator (D-ITG). D-ITG is a tool for generating network traffic. It supports various protocols involved in Wireless Communication. D-ITG allows you to specify traffic patterns, flow rates, packet sizes, and other characteristics. The amount of traffic is enough to congest one AP but not the entire network. The main objective of TiWA is to allocate resources effectively to prevent AP congestion. If the network were entirely congested, the redistribution of user and load would have no impact since all three APs would be congested. Three state-of-the-art approaches, a client driven (CD) approach, a WiMA [3], a Wi-Balance [50] technique were selected to compare TiWA performance. Wi-Balance was chosen because of its effective AP-Level load balancing and the availability of code and WiMA was used as it was a great algorithm, that consider three factors AP load, RSSI, and channel occupancy, to compare with TiWA in which we are using four factors. Using these four factors of TiWA instead of the three factors of WiMA will gives us more efficient results in terms of throughput maximization.

A. EXPERIMENTS

1) EXPERIMENT 1

Experiment 1 aimed to create a typical scenario where densification is used to relieve a congested AP. The objective of TiWA is to redistribute the load to other APs, whenever possible. The focus of this experiment was to observe the impact of the indicator, AP load. To prevent collision domains from affecting each other, channels 1, 6, and 11 were employed. To reduce the impact of mobility on the

TABLE 2. Experiment statistics.

Sr. No.	Channels	Wireless Devices Aspect 1	Wireless Devices Aspect 2	Moving Wireless Devices	Time Duration
1	11,6,1	30	40	NILL	20 Min
2	11,6,1	30	NILL	10	10 Min
3	11,1	30	NILL	NILL	20 Min

experiment and facilitate the interpretation of results, no moving wireless devices were used. The experiment had two aspects, each lasting for 20 minutes. In Aspect 1, 30 wireless devices located in positions illustrated in Figure 3 transmitted. In Aspect 2, an additional 10 wireless devices began transmitting. The wireless devices were positioned in the area that one of the APs covers, allowing load balancing approaches to move the load to nearby APs.

2) EXPERIMENT 2

The objective of Experiment 2 is to measure TiWA responsiveness to variations in wireless device signal quality. In this experiment, two moving wireless devices are used along the route shown in Figure 4. The route was specifically chosen to evaluate the performance of the RSSI and the effects of low signal strength on migrations by ensuring that the wireless devices move inside the ranges of all three APs. The wireless devices travel for 1 minute and then stop at the dots along the course to wait for 2 minutes before moving on. Channels 1, 6, and 11 were utilized, and the experiment lasted for 10 minutes to complete the path. Twenty wireless devices were situated in the same location as the first aspect of the preceding experiment, continuously sending traffic to the AP.

3) EXPERIMENT 3

Experiment 3 is the last one created to evaluate TiWA channel occupancy indicator. The goal of this experiment is to mimic a scene in which densification has already taken place, resulting in the sharing of the collision domain by two APs. TiWA depends heavily on its capacity to address the congestion. TiWA considers channel load rather than AP load. Just channels 1 and 11 were used in this experiment to make two APs share a single collision domain. We tested TiWA to see how it balances AP load, signal strength, and channel occupancy. TiWA redistributed channels when reassociation was unable to relieve congestion. Twenty wireless devices were arranged as illustrated in Figure 5 in order to have a balanced distribution throughout the APs. In order to verify the load distribution over the channels, all wireless devices transmitted continuously for 20 minutes without the usage of any moving wireless devices.

B. RESULTS

By gathering a number of important performance metrics, this portion computes the experiment's outcomes. These indicators include Jain's fairness index, which computes the fairness of resource distribution, and aggregated throughput, which gauges the quantity of relevant information at the application level. It formula is given as:

Jain's Fairness Index formula:

$$\text{Fairness Index} = \frac{\sum(W_i)^2}{x \cdot (\sum W_i)^2}$$

where W_i is the resource distributions. The fairness index formula compares the individual contributions ($(W_i)^2$) to the total resource allocation ($x \cdot (W_i)^2$). Additionally taken into account is the typical frequency of retransmission attempts per transmitted frame, which measures both the quantity and quality of collisions. The average standard deviation of channel occupancy which depicts the variance of occupancy amongst channels employed in the experiment is evaluated alongside channel occupancy, which determines how much channel time is used relative to the overall amount of time. Finally the average physical rate, since it represents network collisions and signal quality chosen by the MCS selection method is calculated.

1) EXPERIMENT 1

Figure 6 presents the results of the Experiment 1, highlighting the superiority of TiWA over the other two tested approaches in both experiment aspects. The improvement is more significant in the second aspect, where the network load increases. The results indicate that better load distribution leads to a more significant improvement in aggregated throughput, as depicted in Figure 6 (a). Specifically, in Aspect 1, TiWA achieves a 7% improvement in aggregated throughput compared to Wi-Balance, 8% improvement when compared to WiMA and a 52% improvement compared to the CD approach. In Aspect 2, the improvement is 12% over Wi-Balance, 10% over WiMA and 58% over the CD approach. Such an improvement in resource allocation is also visible in Jain's Fairness Index shown in Figure 6 (b), where TiWA achieves a 16%, 6% and 27% better fairness in Aspect 1 than Wi-Balance, WiMA and the CD approach, respectively. In Aspect 2, TiWA outperforms aforementioned approaches by 25% in terms of fairness, whereas Wi-Balance shows a similar performance to the CD approach.

The average Frequency of retransmission attempts per frame is presented in Figure 6 (c), indicating that contention-based Wi-Fi medium access results in high frequency of retransmissions due to collisions when channel occupancy is higher. Additionally, signal quality plays a crucial role

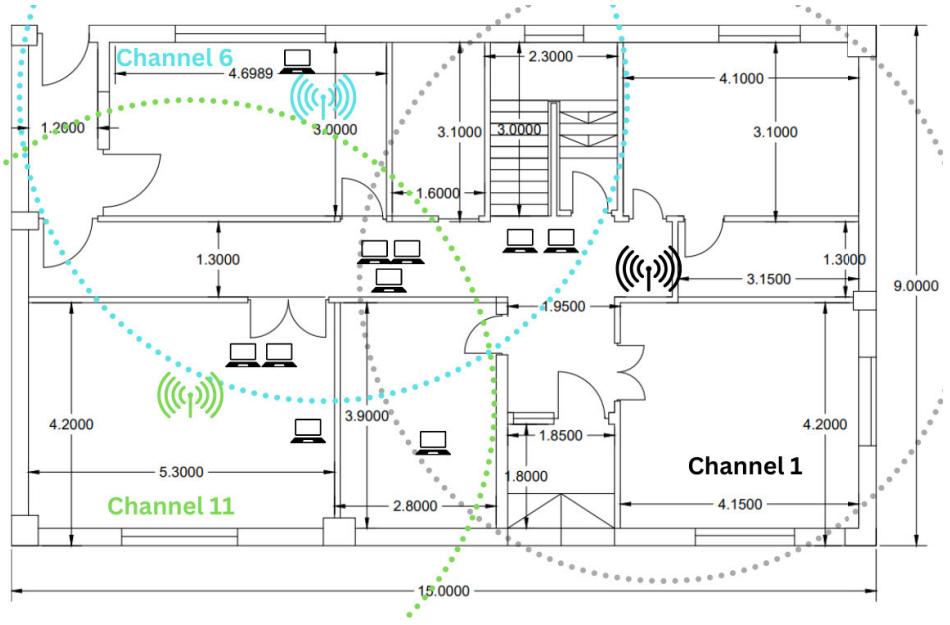


FIGURE 3. Layout of testbed for experiment 1 representing different channels with wireless devices.

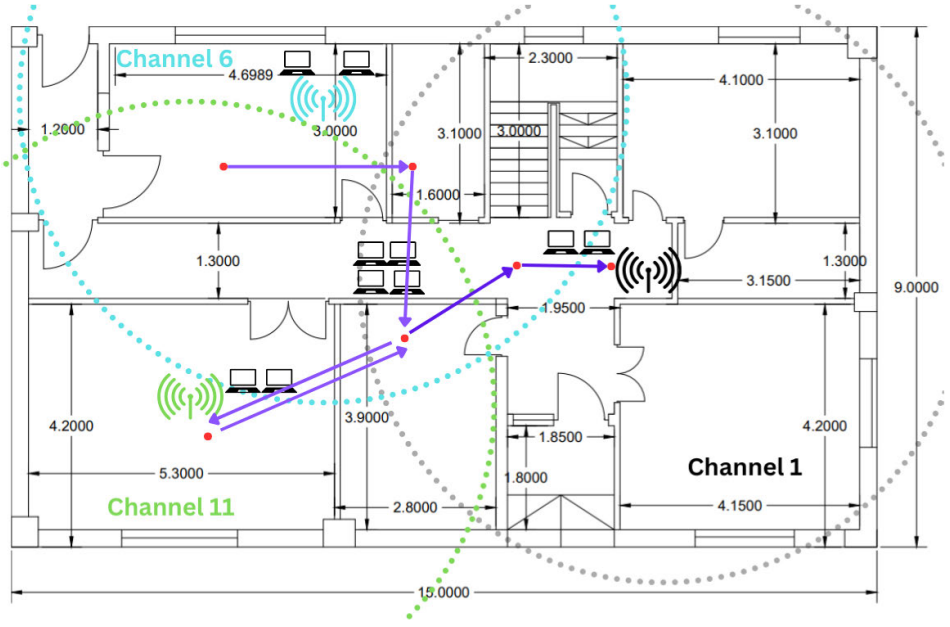


FIGURE 4. Layout of testbed for experiment 2 representing different channels with wireless devices and their path of travelling from one AP to another AP.

in producing retransmissions due to fading and shadowing. By considering these two factors, TiWAs trade-off approach is effective in reducing retransmissions frequency by half. This is consistent with the decrease in channel occupancy shown in Figure 6 (d) and the higher average physical rate depicted in Figure 6 (f). Wi-Balance achieves a more balanced channel usage, but it sacrifices signal quality to do so, as evident by the decrease in the average physical rate. WiMA gives more better results but TiWA surpasses WiMA and Wi-Balance by 19% and 10% respectively in terms of the average physical rate as better signal quality enables the default rate control algorithm, Minstrel, to choose MCS with higher rates and fewer retransmissions.

2) EXPERIMENT 2

The findings of Experiment 2 are presented in Figure 7. The results show that TiWA performs well in adapting to wireless

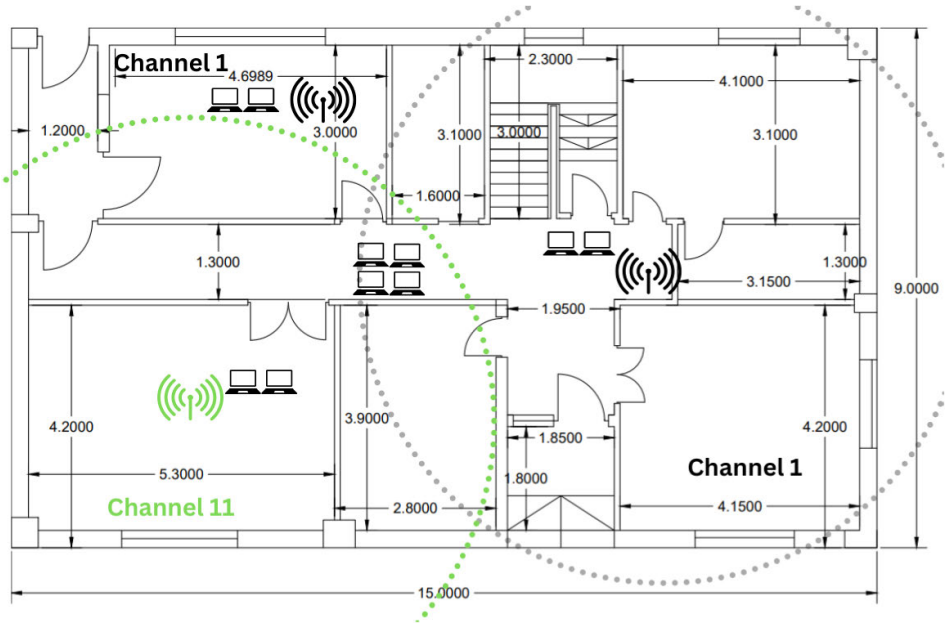


FIGURE 5. Layout of testbed for experiment 3 representing different channels with wireless devices.

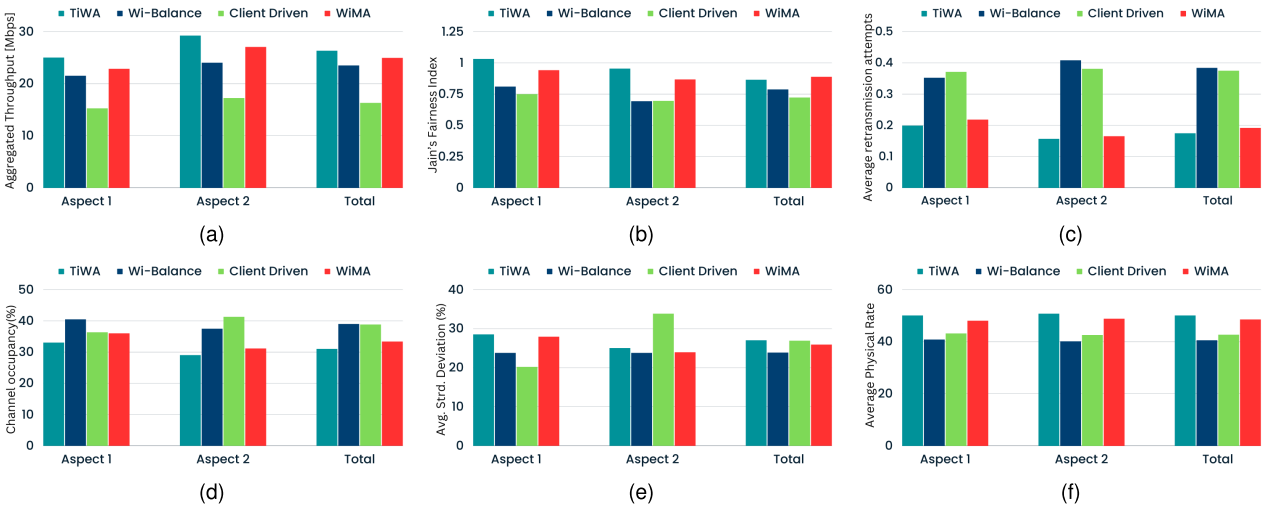


FIGURE 6. Results of Experiment 1 showing Aspects 1,2 and their average as total in (a) Throughput (b) Jain's Fairness Index (c) Number of Retransmissions per frame (d) Channel Occupancy Rate (e) Average Standard Deviation (f) Average Physical Rate.

device movements, leading to improved outcomes. WiMA has also good results but not optimal like TiWA. Conversely, Wi-Balance suffers from the movement of wireless devices and variation in the signal quality, resulting in a negative impact on performance, as depicted in Figure 7 (a). This is because Wi-Balance does not consider the reassociation of the wireless devices to other APs when signal quality degrades. Meanwhile, the CD approaches suboptimal allocation of resources results in congestion and a lower aggregated throughput. In contrast, TiWA strikes a good balance between these two factors, resulting in a 20% improvement in aggregated throughput due to better resource allocation. This improvement in resource allocation also leads to a higher Jain's Fairness Index, as shown in Figure 7 (b), allowing all wireless devices to benefit from the increased throughput.

Retransmissions are shown in Figure 7 (c) and are affected by both collisions and signal quality. Due to the fact that Wi-Balance does not migrate wireless devices using the RSSI, it makes more retransmission attempts than TiWA, WiMA and the CD approach. Figure 7 (d) shows the channel occupancy performance, with Wi-Balance displaying reduced occupancy as a result of wireless device migration to an AP with subpar signal quality, which led to disconnections. TiWA uses channels more equitably than Wi-Balance, WiMA and the CD method, as shown in Figure 7 (e), outperforming by 29%, 11% and 87%. According to the graph, the CD

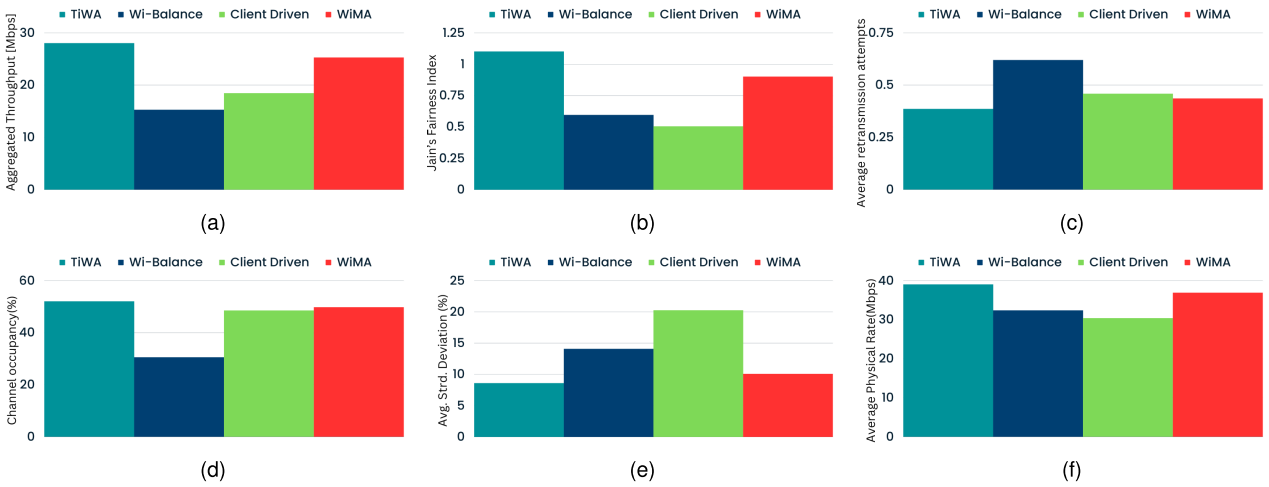


FIGURE 7. Results of Experiment 2 showing Aspects 1,2 and their average as total in (a) Throughput (b) Jain's Fairness Index (c) Number of Retransmissions per frame (d) Channel Occupancy Rate (e) Average Standard Deviation (f) Average Physical Rate.

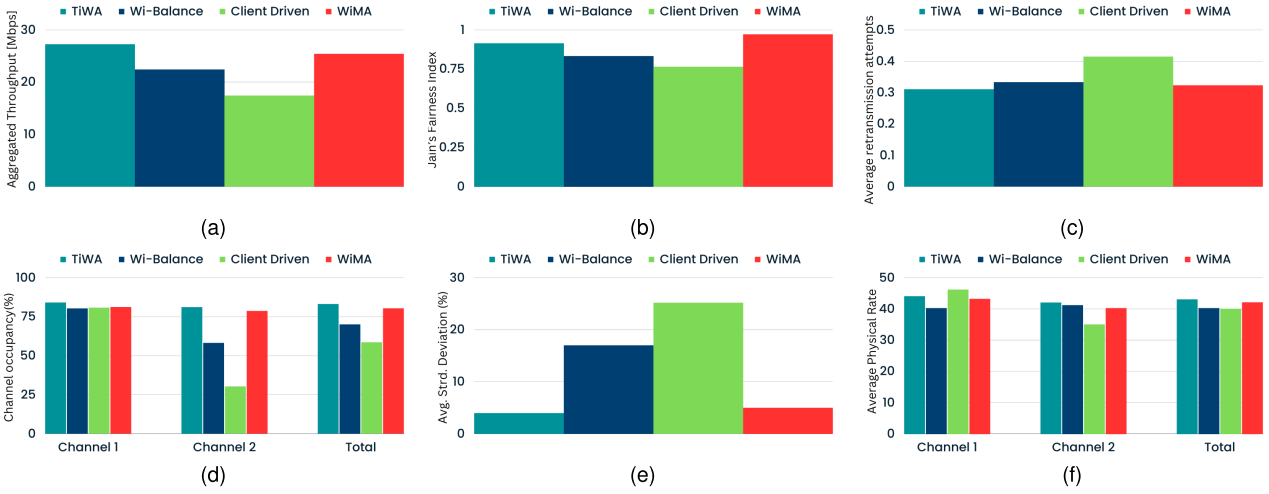


FIGURE 8. Results of Experiment 3 showing Aspects 1,2 and their average as total in (a) Throughput (b) Jain's Fairness Index (c) Number of Retransmissions per frame (d) Channel Occupancy Rate (e) Average Standard Deviation (f) Average Physical Rate.

strategy underutilizes some channels while saturating others. Although TiWA has a worse wireless devices distribution and a lower MCS that was chosen by Minstrel in response to collisions brought by the congestion in the CD approach. Due to the improved performance in accommodating moving wireless devices, TiWA chooses an MCS with a 26% higher physical rate than Wi-Balance, 18% above the CD method and 8% up than WiMA, as illustrated in Figure 7 (f).

3) EXPERIMENT 3

The outcomes of Experiment 3 are depicted in Figure 8 along with the effectiveness of TiWA in establishing good load balancing at the channel level. As demonstrated in Figure 8 (a), TiWA performs better than Wi-Balance, WiMA and the CD method in terms of aggregated throughput, outperforming them by 16%, 10% and 37%, respectively. This improvement in throughput is partially attributable to the more effective use of resources, as shown in Figure 8 (d), where TiWA utilizes channel 11 more effectively while channel 1 is congested by the two APs. The standard deviation of channel usage in Figure 8 (e) demonstrates that TiWA produces a more balanced use of the two channels, whereas Wi-Balance, despite outperforming the CD technique, is less successful at reducing channel occupancy. A more equitable allocation of resources across wireless devices is the outcome of the more balanced utilization, which also raises the Jain's Fairness Index, as seen in Figure 8 (b). TiWA achieves fairness indices that are 10% higher than Wi-Balance, 8% higher than WiMA and 27% higher than the CD method. As shown in Figure 8 (b), TiWA reduces retransmissions by 20% when compared to the CD method, 8% when compared to WiMA and by 4% when compared to Wi-Balance by lowering channel 1 congestion. Congestion is a factor even if signal quality is important in establishing the physical data rate used by

the wireless devices. TiWA, which has a greater average data rate than Wi-Balance, WiMA and the CD method, achieves a trade-off by lowering collisions brought on by congestion while retaining good signal quality, as illustrated in Figure 8 (f). Wi-Balance, in contrast, prevents collisions but disregards quality of signal, whilst the CD technique fails to take the load into account and causes collisions as a result of congestion.

4) RUNNING TIME

Besides the experiments mentioned earlier, the time taken for making a decision has also been assessed. The decision time refers to the duration between the arrival of a report from an AP to the controller and the subsequent action taken, such as migrating a wireless device, redistributing channels, or taking no action. The time of 60 such decisions was measured, and the average decision time was found to be 0.1889 ± 0.088 seconds with a 95% confidence interval. The algorithm assesses a potential redistribution every second, therefore any decision time below 1 second is considered valid for TiWA. Moreover, it is worth noting that since TiWA operates on user and load distributions, it is not strictly bound by time constraints as the factors do not change rapidly.

VI. CONCLUSION

In this paper, we introduced TiWA, a wireless device association strategy for Software-Defined Wi-Fi Networks that aims to balance performance variables and prevent the requirement for densification. Densification is expensive and useless since it invariably results in collision domains, making it ineffectual against congestion when used to sustain performance in the face of rising user and traffic. To delay the need for densification until there is a general and substantial rise in users, TiWA employs a more efficient wireless device association to make better use of wireless resources.

By creating an indicator for each of these measures, TiWA specifically accomplishes a balanced trade-off between packet loss rate, received signal strength indicator (RSSI), throughput, and channel occupancy. Our analysis of real-world performance shows that TiWA enhances the network performance in diverse setups without the need for densification. TiWA increases network aggregated throughput by up to 20%, 39% and 50%, respectively, as compared to Wi-Balance, a cutting-edge system, WiMA and conventional CD techniques. Additionally, TiWA raises Jain's Fairness Index on the wireless device, allowing most wireless devices to take advantage of the improved throughput and allocating resources more efficiently. The optimal allocation reduces congestion-induced retransmissions and poor signal quality while utilizing the unused resources at underloaded APs. As a result, TiWA can support more users and traffic before requiring densification.

Additionally, our findings indicate that TiWA enhances performance even after deployment of densification and many APs are located inside the same collision domain. We hope to implement artificial intelligence (AI) in TiWA, so that the APs make the association decision themselves till a certain load level, in near future.

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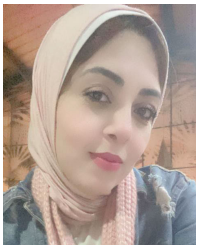
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